

Generation of Jacobi Matrices on the Simplex

d=1

```
In[1]:= ClearAll[P,W,Wab,Wabc,H,a,b,c,h,Ax,Bx,Ay,By,A1,A2,B1,B2,x,y,Jx,Jy,Antmp];
13
14 (* Orthogonal Jacobi polynomials in 1D *)
15 P[a_,b_,n_,x_] :=
16 Assuming[
17
18     Element[n, Integers] &&
19     Element[a, Reals] &&
20     Element[b, Reals] &&
21     Element[x, Reals] &&
22     n ≥ 0 &&
23     a > -1 &&
24     b > -1 &&
25     x ≥ -1 &&
26     x ≤ 1,
27
28     ((-1)^n)/(2^n * n!) *
29     (1-x)^(-a)*(1+x)^(-b) *
30     D[(1-x)^(a+n)*(1+x)^(b+n),{x,n}]
31
32 ];
33
34 (* Pi,Pj are orthogonal in L2([-1,1],W) weighted inner product *)
35 W[a_,b_,x_] :=
36 Assuming[
37
38     Element[a, Reals] &&
39     Element[b, Reals] &&
40     Element[x, Reals] &&
41     a > -1 &&
42     b > -1 &&
43     x ≥ -1 &&
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      x ≤ 1,

      ((1-x)^a) * ((1+x)^b)

];

(* <1,1> = 1 only if we divide inner products by Wab *)
Wab[a_,b_] :=
Assuming[

  Element[a, Reals] &&
13  Element[b, Reals] &&
14  a > -1 &&
15  b > -1,
16
17  Integrate[W[a,b,x],{x,-1,1}]
18
19 ];

20 (* Use H to check orthogonality, and
21 compute structural constants for Pk *)
22 H[a_,b_,i_,j_] :=
23 Assuming[
24
25  Element[a, Reals] &&
26  Element[b, Reals] &&
27  Element[i, Integers] &&
28  Element[j, Integers] &&
29  a > -1 &&
30  b > -1 &&
31  i ≥ 0 &&
32  j ≥ 0,
33
34  Integrate[P[a,b,i,x]*P[a,b,j,x]*W[a,b,x], {x,-1,1}]/Wab[a,b]
35
36 ];
37
38 (* normalize the polynomials *)
39 p[a_,b_,n_,x_] :=
40 Assuming[
41
42  Element[a, Reals] &&
43  Element[b, Reals] &&
44

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        Element[n, Integers] &&
        Element[x, Reals] &&
        a > -1 &&
        b > -1 &&
        n ≥ 0 &&
        x ≥ -1 &&
        x ≤ 1,

        P[a,b,n,x] / Sqrt[H[a,b,n,n]]
    ];

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In[7]:= (* check if we've done everything right so far *)
107 a = 1; b = 3;
108 h[i_,j_] := Integrate[p[a,b,i,x]*p[a,b,j,x]*W[a,b,x],{x,-1,1}]/Wab[a,b];
109 (* expect identity *)
110 MatrixForm[Table[h[i,j],{i,0,5},{j,0,5}]]

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Out[9]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

d = 2

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In[10]:= (* now move on to Tri *)
122 ClearAll[a,b,c,x,y,n,k,Sz]
123 (* Normalization for the measure *)
124 Wabc[a_,b_,c_] :=
125 Assuming[
126
127     Element[a, Reals] &&
128     Element[b, Reals] &&
129     Element[c, Reals] &&
130     a > -1/2 &&
131     b > -1/2 &&
132     c > -1/2,

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Gamma[a+b+c+3/2]/(Gamma[a+1/2]*Gamma[b+1/2]*Gamma[c+1/2])

];
(* Normalized measure *)
W[a_,b_,c_,x_,y_]:=
Assuming[

    Element[a, Reals] &&
    Element[b, Reals] &&
    Element[c, Reals] &&
    Element[x, Reals] &&
    Element[y, Reals] &&
    a > -1/2 &&
    b > -1/2 &&
    c > -1/2 &&
    x ≥ 0 &&
    y ≥ 0 &&
    1-x-y ≥ 0,

    x^(a-1/2)*y^(b-1/2)*(1-x-y)^(c-1/2)*Wabc[a,b,c]

];

(* structure constants *)
H[a_,b_,c_,k_,n_]:=
Assuming[

    Element[a, Reals] &&
    Element[b, Reals] &&
    Element[c, Reals] &&
    Element[k, Integers] &&
    Element[n, Integers] &&
    a > -1/2 &&
    b > -1/2 &&
    c > -1/2 &&
    k ≥ 0 &&
    n ≥ 0,

    Piecewise[
    {{Sqrt[Wabc[a,b,c]/((2*n+a+b+c+1/2)*(2*k+b+c)) *

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Gamma[n+k+b+c+1]*Gamma[n-k+a+1/2]*Gamma[k+b+1/2]*Gamma[k+c+1/2] /
((n-k)!*k!*Gamma[n+k+a+b+c+1/2]*Gamma[k+b+c]), k ≤ n},{0,k>n}}]

];

(* Orthonormal Koornwinder polynomials *)
P[a_,b_,c_,k_,n_,x_,y_]:=
Assuming[

    Element[a, Reals] &&
    Element[b, Reals] &&
    Element[c, Reals] &&
    Element[k, Integers] &&
    Element[n, Integers] &&
    Element[x, Reals] &&
    Element[y, Reals] &&
    a > -1/2 &&
    b > -1/2 &&
    c > -1/2 &&
    k ≥ 0 &&
    n ≥ 0 &&
    x ≥ 0 &&
    y ≥ 0 &&
    1-x-y ≥ 0,

    H[a,b,c,k,n]^(-1) *
    (P[2*k+b+c,a-1/2,n-k,z]/.z→2*x-1) *
    (1-x)^k *
    (P[c-1/2,b-1/2,k,z]/.z→2*y/(1-x)-1)

];

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In[15]:= (* check if we've done everything right so far *)
221
222 (* expect upper trimat*)
223 MatrixForm[Table[H[1/2,1/2,1/2,k,n],{k,0,10},{n,0,10}]]
224 a = 1/2; b = 1/2; c = 1/2 ;
225 h[i_,j_] := Integrate[P[a,b,c,i,i,x,y]*P[a,b,c,j,j,x,y]*W[a,b,c,x,y],{x,0,1},{y,0,1-x}];
226 (* expect identity *)
227 MatrixForm[Table[h[i,j],{i,0,5},{j,0,5}]]

```

Out[15]//MatrixForm=

$$\begin{pmatrix} 1 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{2} & \frac{1}{\sqrt{5}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{7}} & \frac{1}{2\sqrt{2}} & \frac{1}{3} & \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{11}} \\ 0 & \frac{1}{\sqrt{6}} & \frac{1}{3} & \frac{1}{2\sqrt{3}} & \frac{1}{\sqrt{15}} & \frac{1}{3\sqrt{2}} & \frac{1}{\sqrt{21}} & \frac{1}{2\sqrt{6}} & \frac{1}{3\sqrt{3}} & \frac{1}{\sqrt{30}} & \frac{1}{\sqrt{33}} \\ 0 & 0 & \frac{1}{\sqrt{15}} & \frac{1}{2\sqrt{5}} & \frac{1}{5} & \frac{1}{\sqrt{30}} & \frac{1}{\sqrt{35}} & \frac{1}{2\sqrt{10}} & \frac{1}{3\sqrt{5}} & \frac{1}{5\sqrt{2}} & \frac{1}{\sqrt{55}} \\ 0 & 0 & 0 & \frac{1}{2\sqrt{7}} & \frac{1}{\sqrt{35}} & \frac{1}{\sqrt{42}} & \frac{1}{7} & \frac{1}{2\sqrt{14}} & \frac{1}{3\sqrt{7}} & \frac{1}{\sqrt{70}} & \frac{1}{\sqrt{77}} \\ 0 & 0 & 0 & 0 & \frac{1}{3\sqrt{5}} & \frac{1}{3\sqrt{6}} & \frac{1}{3\sqrt{7}} & \frac{1}{6\sqrt{2}} & \frac{1}{9} & \frac{1}{3\sqrt{10}} & \frac{1}{3\sqrt{11}} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{66}} & \frac{1}{\sqrt{77}} & \frac{1}{2\sqrt{22}} & \frac{1}{3\sqrt{11}} & \frac{1}{\sqrt{110}} & \frac{1}{11} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{91}} & \frac{1}{2\sqrt{26}} & \frac{1}{3\sqrt{13}} & \frac{1}{\sqrt{130}} & \frac{1}{\sqrt{143}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2\sqrt{30}} & \frac{1}{3\sqrt{15}} & \frac{1}{5\sqrt{6}} & \frac{1}{\sqrt{165}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3\sqrt{17}} & \frac{1}{\sqrt{170}} & \frac{1}{\sqrt{187}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{190}} & \frac{1}{\sqrt{209}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{231}} \end{pmatrix}$$

Out[18]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

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In[19]:= (* our lex order is (x^n,x^n-1y,..,y^n) *)
231 ClearAll[a,b,c,x,y,n,nn,k,Sz,Jn1,Jn2,B1,B2,inds,inds1,An1,An2,Bn1,Bn2,A1,A2]
232
233 P[a_,b_,c_,n_,x_,y_]:=
234 Assuming[
235
236     Element[a, Reals] &&
237     Element[b, Reals] &&
238     Element[c, Reals] &&
239     Element[n, Integers] &&
240     Element[x, Reals] &&
241     Element[y, Reals] &&

```

```

a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0 &&
x ≥ 0 &&
y ≥ 0 &&
1-x-y ≥ 0,

Table[P[a,b,c,k,n,x,y],{k,0,n}]

```

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]
```

```
(* Blocks of Jacobi matrix*)
```

```
Ax[a_,b_,c_,n_] :=
```

```
Assuming[
```

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0,

```

```

Table[Integrate[Outer[Times,
x*P[a,b,c,k,x,y],
P[a,b,c,k+1,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x},{k,0,n-2}]

```

```
];
```

```
Ay[a_,b_,c_,n_] :=
```

```
Assuming[
```

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0,

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Table[Integrate[Outer[Times,
y*P[a,b,c,k,x,y],
P[a,b,c,k+1,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x}},{k,0,n-2}]

];

```

```

Bx[a_,b_,c_,n_] :=
Assuming[

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
x*P[a,b,c,k,x,y],
P[a,b,c,k,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x}},{k,0,n-1}]

];

```

```

By[a_,b_,c_,n_] :=
Assuming[

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
y*P[a,b,c,k,x,y],
P[a,b,c,k,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x}},{k,0,n-1}]

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```
];
```

```
In[25]:= (* generate block recurrence matrices *)
```

```
335 a = 1/2; b = 1/2; c = 1/2; n = 5;
```

```
336 B1 = Bx[a,b,c,n];
```

```
337 A1 = Ax[a,b,c,n];
```

```
338 B2 = By[a,b,c,n];
```

```
339 A2 = Ay[a,b,c,n];
```

```
340
```

```
In[30]:= (* assemble into Jacobi matrices for each dim *)
```

```
344 n = 5;
```

```
345 Sz = n *(n+1)/2;
```

```
346 Jn1 = ConstantArray[0,{Sz,Sz}];
```

```
347 Jn2 = ConstantArray[0,{Sz,Sz}];
```

```
348 inds = {1,1};
```

```
349 For [nn = 1, nn ≤ n-1, nn++,
```

```
350 Jn1[[inds[[1]];;inds[[2]],inds[[1]];;inds[[2]]] = B1[[nn];
```

```
351 Jn2[[inds[[1]];;inds[[2]],inds[[1]];;inds[[2]]] = B2[[nn];
```

```
352 inds1 = inds + {nn,nn+1};
```

```
353 Jn1[[inds1[[1]];;inds1[[2]],inds1[[1]];;inds1[[2]]] = A1[[nn];
```

```
354 Jn1[[inds1[[1]];;inds1[[2]],inds[[1]];;inds[[2]]] = Transpose[A1[[nn];
```

```
355 Jn2[[inds[[1]];;inds[[2]],inds1[[1]];;inds1[[2]]] = A2[[nn];
```

```
356 Jn2[[inds1[[1]];;inds1[[2]],inds[[1]];;inds[[2]]] = Transpose[A2[[nn];
```

```
357 inds = inds + {nn,nn+1}
```

```
358 ];
```

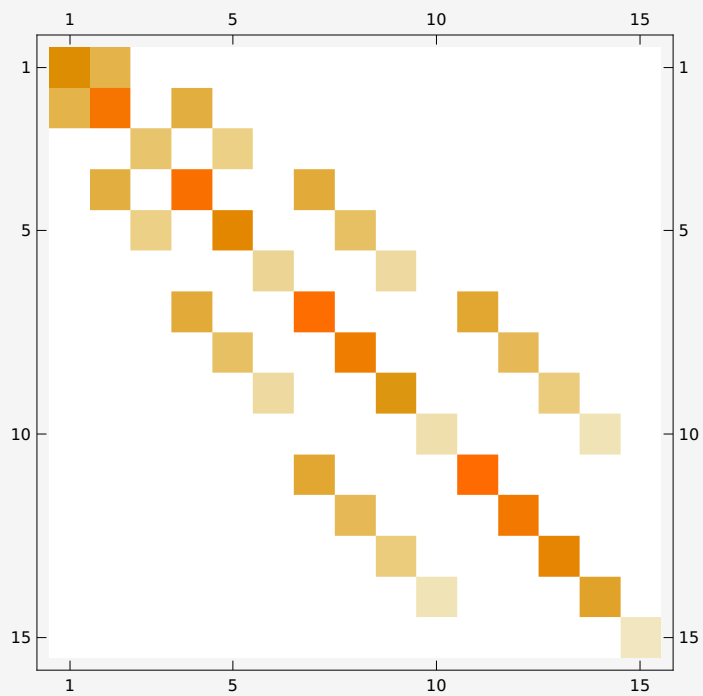
```
359 Jn1[[inds[[1]];;inds[[2]],inds[[1]];;inds[[2]]] = B1[[n];
```

```
360 Jn2[[inds[[1]];;inds[[2]],inds[[1]];;inds[[2]]] = B2[[n];
```

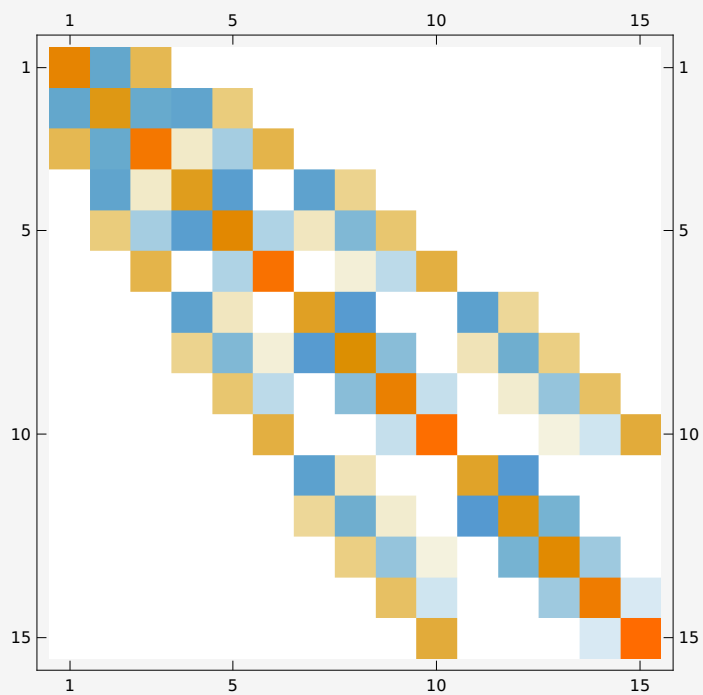
```
In[38]:=MatrixPlot[Jn1]
```

```
MatrixPlot[Jn2]
```

```
Out[38]=
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```
Out[39]=
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